

Periodic Classification of Elements

3



Reference: Coursebook - IIT Foundation Chemistry Class 9; Chapter - Periodic Classification of Elements; Page number - 3.1–3.13

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

- The electronic configuration of element 'X' is 2, 8 and 7. Identify the true statement regarding X.
 - It belongs to VII-A group and it is a typical element.
 - It is a halogen and it is a bridge element.
 - It belongs to the fourth period and VI-A group.
 - It is a chalcogen and it belongs to the third period.
- Identify the best oxidizing agent among the following.
 - K
 - O
 - F
 - N
- Identify the set of three elements which can obey the Law of triads.
 - Na, K, Rb
 - Be, Mg, Ca
 - F, Cl, Br
 - C, Si, Ge
- Arrange the elements mentioned below in the order of maximum:
 - Metallic character
 - Electronegativity
 - Ionization energy
 - Electron affinity
 - A belongs to the first period and the 18th group.
 - B belongs to the sixth period and the first group.
 - C belongs to the third period and the 17th group.
 - D belongs to the second period and 17th group.
 - BDAC
 - BDCA
 - DBAC
 - DBCA
- 'A', 'B', 'C' and 'D' are four elements belonging to the third period. A and B can form monovalent and divalent cations, respectively. C and D can form monovalent and divalent anions, respectively. Arrange the corresponding elements in the increasing order of sizes of the respective ions.
 - BADC
 - ABCD
 - CDBA
 - ABDC

6. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Caesium	(A) Second long period
(ii) Xenon	(B) Very long period
(iii) Potassium	(C) Incomplete period
(iv) Radium	(D) Long period

- (a) (i) → (B); (ii) → (C); (iii) → (D); (iv) → (A)
 (b) (i) → (B); (ii) → (A); (iii) → (D); (iv) → (C)
 (c) (i) → (D); (ii) → (C); (iii) → (A); (iv) → (B)
 (d) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (C)

7. **Assertion (A):** The order of the first ionization potential (I.P.) values is $\text{Mg} > \text{Al} > \text{Na}$.

Reason (R): Across the period, I.P. values generally increase.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

8. **Assertion (A):** In a given period, halogens possess the highest electron affinity (E.A.) values.

Reason (R): Across the period, nuclear charge increases and hence, atomic size decreases.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

9. Two metals 'A' and 'B' showed drastic increase in 2nd and 4th I.P. values, respectively. Identify the atomic numbers of A and B.

- (a) 19, 13 (b) 11, 20 (c) 12, 31 (d) 37, 19

10. If the difference in the number of electrons present in the valence shell and penultimate shell of an element 'X' is four, then X belongs to _____.

- (a) 3rd period, V-A group (b) 4th period, III-A group
 (c) 3rd period, IV-A group (d) 5th period, III-A group

11. The atomic numbers of five elements 'A', 'B', 'C', 'D' and 'E' are 19, 16, 11, 7 and 38, respectively. The order of atomic volumes of the elements is _____.

- (a) $D > C > B > A > E$ (b) $E > A > C > B > D$
 (c) $E > D > C > B > A$ (d) $A > E > C > B > D$

Space for rough work

Space for rough work

12. Atoms of two elements 'A' and 'B' have 56 and 37 protons, respectively. Identify the incorrect statement among the following regarding A and B.
- (a) Both A and B are metals.
 - (b) A and B belong to different groups and different periods.
 - (c) A is alkali metal B is alkaline earth metal.
 - (d) Both A and B can form stable cations.
13. The order of ionic sizes of Na^+ , Mg^{+2} , F^- and O^{-2} is _____.
- (a) $\text{Mg}^{+2} > \text{Na}^+ > \text{O}^{-2} > \text{F}^-$
 - (b) $\text{O}^{-2} > \text{F}^- > \text{Mg}^{+2} > \text{Na}^+$
 - (c) $\text{Mg}^{+2} > \text{O}^{-2} > \text{Na}^+ > \text{F}^-$
 - (d) $\text{O}^{-2} > \text{F}^- > \text{Na}^+ > \text{Mg}^{+2}$
14. The electronic configurations of three elements 'A', 'B' and 'C' are 2, 8, 3; 2, 8, 18, 4; 2, 8, 4, respectively. The order of the non-metallic character is _____.
- (a) $\text{C} > \text{A} > \text{B}$
 - (b) $\text{B} > \text{C} > \text{A}$
 - (c) $\text{C} > \text{B} > \text{A}$
 - (d) $\text{B} > \text{A} > \text{C}$
15. Element 'X' undergoes sublimation on heating. Identify the group to which X belongs.
- (a) IV-A
 - (b) V-A
 - (c) VI-A
 - (d) VII-A

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Identify the electronic configuration of an element which belongs to the second long period in the periodic table.

(a) 2, 8, 18, 2 (b) 2, 8, 18, 8
(c) 2, 8, 18, 8, 2 (d) 2, 8, 8

2. The reducing property is maximum for _____.

(a) K (b) Ca (c) Cs (d) F

3. Identify the set of atomic numbers which does not obey the Döbereiner's Law of triads.

(a) 20, 38, 56 (b) 3, 11, 19
(c) 17, 35, 53 (d) 5, 13, 31

4. The electron structures of four elements are given below.

(i) A $\rightarrow$ 2, 8, 1 (ii) B $\rightarrow$ 2, 8, 8
(iii) C $\rightarrow$ 2, 8, 7 (iv) D $\rightarrow$ 2, 8, 18, 8, 1

Arrange the above elements in the order of elements with:

- (A) zero electron affinity.
(B) minimum ionization energy.
(C) good oxidizing agent.
(D) best reducing agent in the third period.

(a) DBAC (b) BDAC
(c) DBCA (d) BDCA

5. Study the following statements:

- (A) Element 'A' requires the least energy to form cation.
(B) Element 'B' requires more energy to form stable cation.
(C) Element 'C' gives out more energy during the formation of anion.
(D) Element 'D' gives out less energy during the formation of anion.

The given elements belong to the same period. Arrange the elements in the increasing order of their atomic sizes.

(a) ABCD (b) CDBA
(c) BACD (d) BADC

6. Match the entries of Column A with those of Column B.

Column A (Electronic Configuration)	Column B (Type of Element)
(i) 2, 8, 8, 1	(A) Chalcogen
(ii) 2, 8, 18, 8, 2	(B) Halogen
(iii) 2, 8, 18, 18, 7	(C) Alkaline earth metal
(iv) 2, 8, 18, 6	(D) Alkali metal

Space for rough work

- (a) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
 (b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)
 (c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
 (d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)

7. **Assertion (A):** The second I.P of Na is greater than that of Mg.

Reason (R): Na^+ ion has a stable octet configuration.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

8. **Assertion (A):** Formation of F^- is an exothermic process and that of F^{2-} is an endothermic process.

Reason (R): Repulsive force is directly proportional to the number of electrons in the species.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

9. The order of the first I.P. values of three elements 'X', 'Y' and 'Z' is $Y > Z > X$. What could be the elements X, Y and Z?

- (a) Al, Si, P (b) C, N, O
 (c) N, O, F (d) S, Cl, Ar

10. Element 'X' belongs to the fourth period and VI-A group. The difference in the number of electrons present between the last and first shell is _____.

- (a) 2 (b) 4 (c) 6 (d) 1

11. Identify the correct order of atomic sizes of the following elements.

- (a) $\text{O} > \text{F} > \text{Ne}$ (b) $\text{Ar} > \text{S} > \text{Cl}$
 (c) $\text{Be} > \text{Cl} > \text{S}$ (d) $\text{Na} > \text{K} > \text{Ca}$

12. 'X', 'Y' and 'Z' belong to II-A group and obey the Döbereiner's Law of triads. Identify the true statement regarding X, Y and Z.

- (a) $X \rightarrow \text{Ca}$, $Y \rightarrow \text{Sr}$, $Z \rightarrow \text{Ba}$, and Ba is a good reducing agent.
 (b) $X \rightarrow \text{Mg}$, $Y \rightarrow \text{Ca}$, $Z \rightarrow \text{Sr}$, and Ca is a good reducing agent.
 (c) $X \rightarrow \text{Mg}$, $Y \rightarrow \text{Ca}$, $Z \rightarrow \text{Sr}$, and Mg has low I.P value.
 (d) $X \rightarrow \text{Ca}$, $Y \rightarrow \text{Sr}$, $Z \rightarrow \text{Ba}$, and Mg has lower I.P value than that of X, Y and Z.

13. Identify the correct order of sizes of neutral atoms and their corresponding ions.

- (a) $A > A^{+1}$, $B^{2-} > B$ (b) $A > A^+$, $B > B^{2-}$
 (c) $A^{+1} > A$, $B^{2-} > B$ (d) $A^{+1} > A$, $B > B^{2-}$

3.6 Chapter 3 Periodic Classification of Elements

14. The electronic configuration of element 'X' is 2, 8 and 6. X is surrounded by four elements as shown in the figure. Identify the element with the least non-metallic character.

	B	
A	X	C
	D	

- (a) D (b) C (c) B (d) A
15. Element 'A' is used for manufacture of match sticks. Predict the position of A in the modern periodic table.
- (a) V-A group and third period
(b) IV-A group and third period
(c) V-A group and fourth period
(d) IV-A group and fourth period

Space for rough work

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

- Based on the increasing order of which characteristic of the elements, Mendeleev's periodic table is designed?
(a) Atomic weight (b) Atomic size
(c) Atomic number (d) Number of core electrons
- In a Döbereiner's triad, the atomic weights of first two elements are x and y . What should be the atomic weight of the 3rd element?
(a) $x + y$ (b) $x - y$ (c) $x + 2y$ (d) $2y - x$
- The atomic number of an element is 30. Determine its position in the periodic table.
(a) 3rd period, 12th group
(b) 4th period, 12th group
(c) 4th period, 14th group
(d) 5th period, 10th group
- Which of the following type of elements is associated with the highest density?
(a) Alkali metal
(b) Alkaline earth metal
(c) Transition metal
(d) Halogen
- Which of the following elements is the most electropositive?
(a) Li (b) Be (c) B (d) C
- Arrange the following elements in the ascending order of the group numbers to which they belong. (Consider the group numbers from 1 to 18.)

Element	Atomic Number
A	19
B	15
C	16
D	9

- (a) ABCD (b) DCBA (c) CDBA (d) BDCA
- The atomic size of an element is _____.
(a) directly proportional to the nuclear charge
(b) directly proportional to the number of valence electrons
(c) inversely proportional to the number of core electrons
(d) inversely proportional to the effective nuclear charge

8. **Assertion (A):** Ionization energies of the representative elements generally increase along a period.

Reason (R): Differentiating electrons enter the same shell of the representative elements along a period.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
9. Which of the following is the best oxidizing agent?
 (a) F (b) O (c) C (d) Mg
10. Which of the following has the highest non-metallic character?
 (a) Cl (b) S (c) Br (d) P
11. The order of successive ionization energies x , y and z of an element is $x < y < z$. What could be the number of valence electrons in the atom of the element?
 (a) 1 (b) 2 (c) 3 (d) 4
12. Which of the following has the largest radius?
 (a) X (b) X^{+1} (c) X^{+2} (d) X^{+3}
13. 'X' is a good reducing agent. Which of the following inferences regarding X is not correct?
 (a) It is a metal.
 (b) Its electropositivity is high.
 (c) Its electron affinity is low.
 (d) It has a tendency to gain electrons.
14. An element belongs to group 4, considering the group numbers as 1 to 18. It is a/an _____.
 (a) alkali metal (b) alkaline earth metal
 (c) transition metal (d) metalloid

15. Match the statements of Column A with those of Column B.

Column A (Description of an Element)	Column B (Electronic Configuration)
(i) Element with the highest electron affinity	(A) 2, 8, 8, 1
(ii) Noble gas belonging to the fourth period	(B) 2, 8, 18, 1
(iii) Element with the largest atomic size in the fourth period	(C) 2, 8, 7
(iv) Metal belonging to the fourth period and group 11	(D) 2, 8, 18, 8

- (a) $(i) \rightarrow (C)$; $(ii) \rightarrow (D)$; $(iii) \rightarrow (B)$; $(iv) \rightarrow (A)$
(b) $(i) \rightarrow (C)$; $(ii) \rightarrow (D)$; $(iii) \rightarrow (A)$; $(iv) \rightarrow (B)$
(c) $(i) \rightarrow (D)$; $(ii) \rightarrow (C)$; $(iii) \rightarrow (B)$; $(iv) \rightarrow (A)$
(d) $(i) \rightarrow (D)$; $(ii) \rightarrow (A)$; $(iii) \rightarrow (B)$; $(iv) \rightarrow (C)$

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

- Which property of elements is considered as the most fundamental property in the construction of the modern periodic table?
 - Atomic weight
 - Atomic number
 - Density of the atom
 - Number of valence electrons
- In a Döbereiner's triad, atomic weights of the first, second and third elements are x , y and z , respectively. What is the relation among x , y and z ?
 - $x = \frac{Y+Z}{2}$
 - $2y = x + z$
 - $z = \frac{x-y}{2}$
 - $y = \frac{x+z}{2}$
- An element belongs to the 4th period and the 15th group. Determine its atomic number.
 - 15
 - 32
 - 33
 - 31
- Identify the element with the least density.
 - K
 - Ca
 - Fe
 - Na
- The electropositivity of an element is pertinent to its _____.
 - tendency of gaining an electron
 - tendency of losing an electron
 - tendency of pulling shared pair of electrons
 - tendency of losing a nucleon
- Which of the following elements has the least group number?
 - Element: P $\rightarrow$ Atomic number is 13.
 - Element: Q $\rightarrow$ Atomic number is 20.
 - Element: R $\rightarrow$ Atomic number is 6.
 - Element: S $\rightarrow$ Atomic number is 14.
- Arrange the following elements in decreasing order of atomic size.

Element: 'A' $\rightarrow$ Atomic number is 7.

Element: 'B' $\rightarrow$ Atomic number is 3.

Element: 'C' $\rightarrow$ Atomic number is 9.

Element: 'D' $\rightarrow$ Atomic number is 11.

 - CDBA
 - DBAC
 - ACBC
 - BDAC